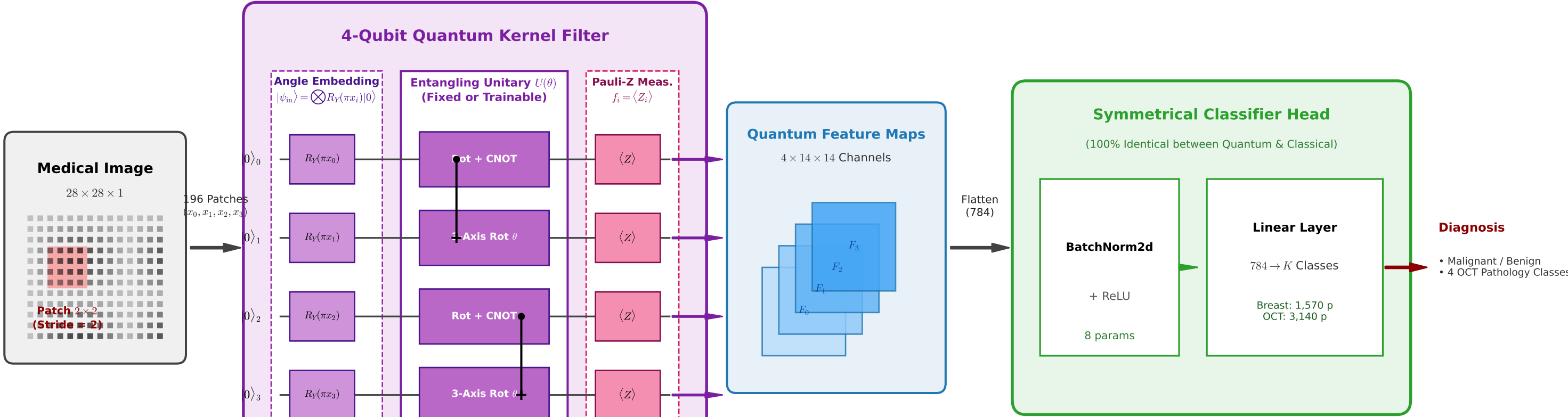


Figure 1: End-to-End Architecture of the Quanvolutional Neural Network (Quanvolution)

Comparison of Symmetrical 4-Qubit Quantum Kernel vs. Minimum Classical Convolutional Filter



Classical Baseline (Symmetrical Minimum CNN):

Conv2D($c_{in} = 1, c_{out} = 4, k = 2, s = 2$) (20 params)

→ 4 × 14 × 14 Feature Maps

→ Same Classifier Head

★ Key Fairness Guarantee: Same parameter count in head, same feature dimensions (784), strictly isolated feature extractor.